ES-125 Egg Drop Presentation
M.O.D.E
"Motion Of Dropping Eggs "

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Design Constraints and Considerations

Design Considerations	
System Height	< 0.5 m
System Mass	< 2 kg
Maximum Acceleration	< 100 m/s^2
Drop Height	≤ 6 m
Impact Force on the Egg	< 40 N

Design Parameters	
Damper Maximum Stroke	0.03302 m
Carriage Mass	1.05 kg
Spring Constant Range (per spring)	437 – 669 N/m
Damping Coefficient Range (per damper)	48 - 135 N/(ms)

Optimization for Constant Acceleration

For our deceleration, we imposed the condition that the acceleration for our system would be constant at 300 m/s^2. Given this optimization, we developed equations to solve for a curve that the damper would act along such that this constant acceleration of the mass would be maintained for all time t.

Geometry Equations:

- $F = -c\left(\frac{dx}{dt}\right) = -c\left(\frac{dx}{dy}\right)\left(\frac{dy}{dt}\right)$
- $tan\Theta = \frac{dy}{dx}$
- $vo = \sqrt{2gh}$
- $y(t) = at^2 - vo$
- $t = vo\sqrt{vo^2 + 2ay}$
- Add in a heavy-side spring to help ease velocity to zero at the end of the masses travel along the rail

Parametrization:

- $F = -c\left(\frac{dy}{dt}\right)\left(\frac{dx}{dy}\right)\left(\frac{\frac{dx}{dy}}{(1+\left(\frac{dx}{dy}\right)^2}\right)$
- $-\frac{ma}{c} = \frac{\frac{dx}{dy}}{(1+\left(\frac{dx}{dy}\right)^2}$
- $-\frac{ma}{c} = \frac{\frac{dx}{dy}}{(1+\left(\frac{dx}{dy}\right)^2}$
- $(\frac{dy}{dx})^2 = -\frac{c\left(\frac{dy}{dt}\right)}{ma} - 1$
- $\frac{dy}{dx} = \sqrt{\frac{c\sqrt{(vo^2+2ay)}}{ma} - 1}$

Equivalent System and Equations of Motion

- We considered a 1 DOF system that had a damping force that varied over time based on the curve, y(x) that we compressed the damper with.
- Our generalized coordinate for our EOM is the y-direction.

$$x = x(y)$$

$$\frac{dx}{dt} = \frac{dx}{dy}\frac{dy}{dt}$$

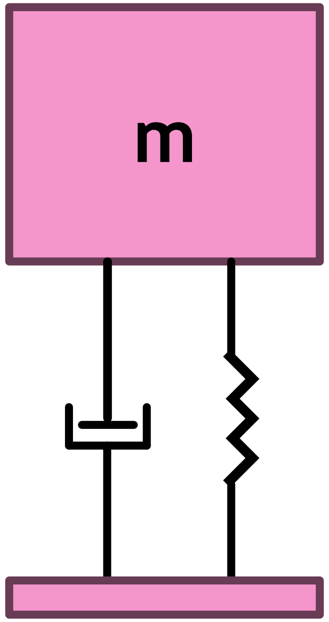
$$Kinetic\ Energy: T = \frac{1}{2}m(\frac{dy}{dx})^2$$

$$Potential\ Energy: P = mgh$$

$$Dissipation: D = \frac{1}{2}C(\frac{dy}{dx}\frac{dy}{dt})^2$$

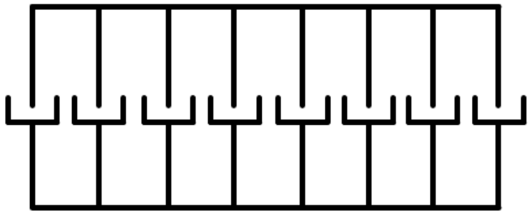
$$Lagrange: \frac{d}{dt}\left(\frac{dT}{d\dot{y}}\right) + \frac{dD}{d\dot{x}} - \frac{dP}{dy}$$

$$EOM: m\left(\frac{d^2y}{dt^2}\right) - mg - c\left(\frac{dx}{dy}\right)^2 (\frac{dy}{dt})$$



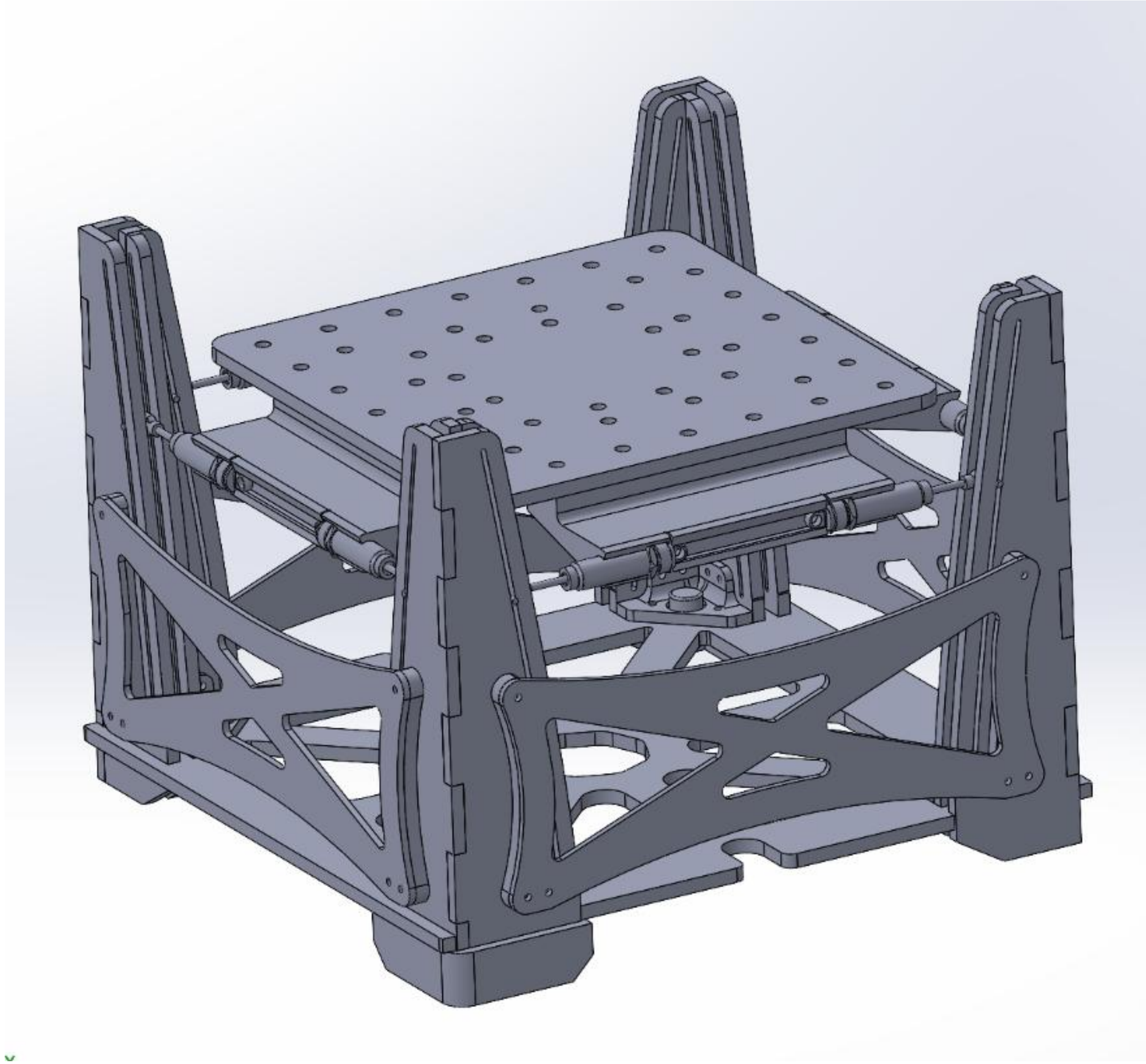
Equivalent Damping and Spring Stiffness

- Equivalent Damping: Dampers in Parallel
- $C_{total} = \sum c_n$
- $C_{total} = 8(135) = 1080$ N/(m–s)



- Equivalent Spring Stiffness: We included no springs in our system
- $K_{total} = \sum k_n$
- $K = 0\ N/m$

CAD Model



System Expectations

- Our model parameterization considered the optimal path for the damper compression to follow considering the drop height of 6 meters, which therefore caused an initial velocity of 10.8499 m/s^2. Because of this optimization, we are hopefully that the system will be able to minimize the maximum acceleration of the system by ensuring that the deceleration of the system is constant at 300 m/s^2.
- In our testing we found that our system was problematic in that it did not account for friction on the rails, so as our system is not frictionless as assumed in the model, there are expected problems in the systems ability to maintain constant acceleration.